Computational Predictive Model for Sustainable Optimization of Cocoa Harvesting by using Advanced Technological Tools

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Introduction

Biodiversity

https://pbs.twimg.com/media/FFXXGIdX0AEMVNq?format=jpg&name=4096x4096

Cocoa crop

Cacao for peace
Cacao for Development

4.7 million tons of dry cocoa (ICCO)

Main problem

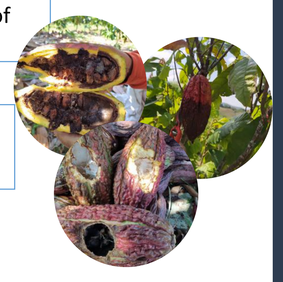
- Non-optimal harvesting strategies
- Untimely harvest
- Crop losses
- Decrease in quality

Empirical and subjective harvesting strategies

Influence of the climatic factors

Loss of 50% of production

Organoleptic characteristics are reduced

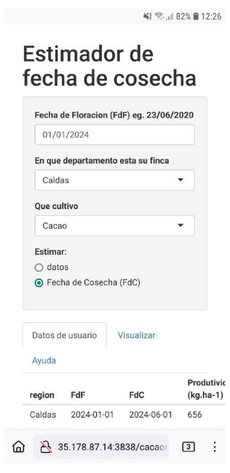


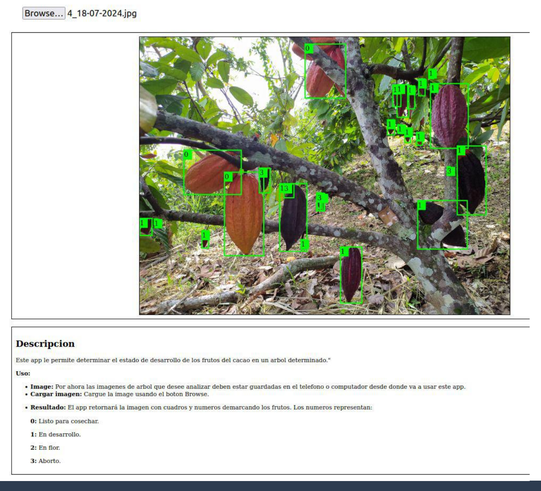
Results

Image repository

Classification Performance

App Development





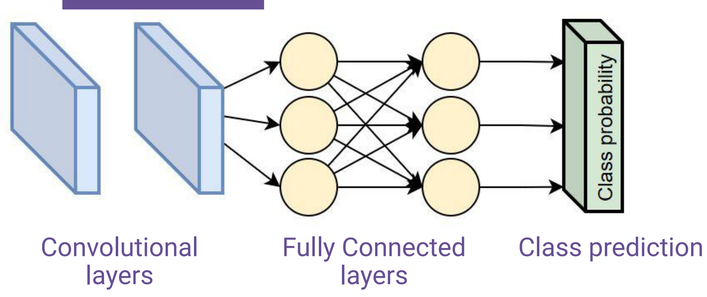
Methodology

Cocoa production chain - Images from flower to harvest

Data preparation using CVAT (Krenzer et al., 2022)

Class
A. Flower
B. Progress
C. Ripe
D. Abortion

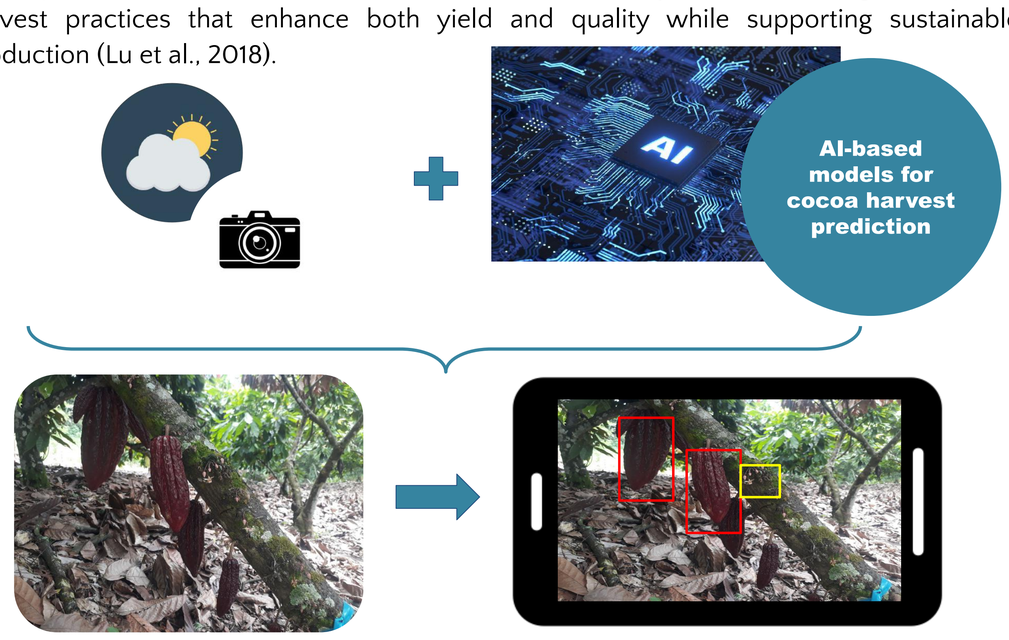
YOLO V8.0



YOLO architecture adapted from (Diwan et al., 2023)

Conclusion

Cocoa production faces various challenges, including climate variability, multiple pathologies, and the complexities of determining the optimal harvest time (Heredia-Gómez et al., 2020). This study proposes a computational predictive model that combines climate data with advanced image analysis techniques to address these issues. The model provides a comprehensive solution focused on the cocoa industry's needs, aiming to improve harvest practices that enhance both yield and quality while supporting sustainable production (Lu et al., 2018).



Acknowledgments

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